

Mental Health Screening Report

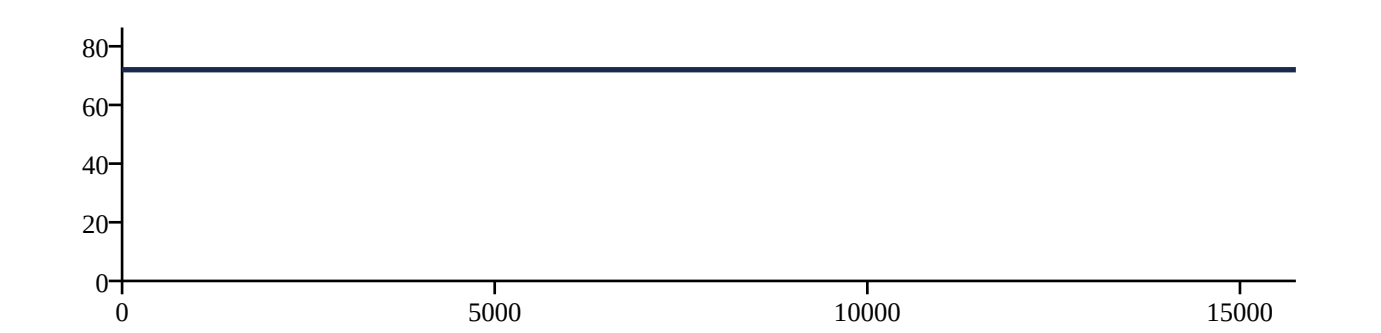
Bio-Synchronous Stress Mapping System

Session Timestamp:	September 25, 2026 - 11:36:23	Subject ID:	SUBJ_WESAD_01 (Screening Protocol)
Total Duration:	1051 min 0 sec	Analysis Window:	30-second Rolling (15s Step, 50% Overlap)

1. Session Summary

Metric Description	Observed Value
Total 30-Second Rolling Windows Analyzed	1051
Stress Windows Detected	0
Non-Stress Windows Detected	1051
Overall Session Stress Percentage	0.0%
Mean Heart Rate (HR)	N/A
Mean HRV (RMSSD)	N/A
Mean Electrodermal Activity (EDA)	N/A
Peak Model Confidence Score	78.7%

2. Heart Rate (HR) Timeline



3. Electrodermal Activity (EDA) Timeline



4. Stress Classification Timeline per 30s Window (50% Overlap)

W1:		NON-STRESS (72.5%)
W2:		NON-STRESS (72.5%)
W3:		NON-STRESS (72.5%)
W4:		NON-STRESS (72.5%)
W5:		NON-STRESS (72.5%)
W6:		NON-STRESS (72.5%)
W7:		NON-STRESS (72.5%)
W8:		NON-STRESS (72.5%)
W9:		NON-STRESS (72.5%)
W10:		NON-STRESS (72.5%)
W11:		NON-STRESS (72.5%)
W12:		NON-STRESS (72.5%)
W13:		NON-STRESS (72.5%)
W14:		NON-STRESS (72.5%)
W15:		NON-STRESS (72.5%)
W16:		NON-STRESS (72.5%)
W17:		NON-STRESS (72.5%)
W18:		NON-STRESS (72.5%)
W19:		NON-STRESS (72.5%)
W20:		NON-STRESS (72.5%)
W21:		NON-STRESS (72.5%)
W22:		NON-STRESS (72.5%)
W23:		NON-STRESS (72.5%)
W24:		NON-STRESS (72.5%)
W25:		NON-STRESS (72.5%)

5. Non-Clinical Decision Support & Interpretation

The subject showed predominantly baseline physiological responses during the session (100.0% of analyzed 30-second rolling windows classified as NON-STRESS). Bio-signal parameters remained within normal resting limits throughout data collection.

Non-Clinical Decision Support Disclaimer: This report summarizes model-estimated physiological stress responses and signal characteristics during the recorded session. It is intended for research and non-clinical decision support and does not constitute a medical or psychiatric diagnosis.

6. Technical Appendix

System Parameter	Specification / Metric Value
Machine Learning Model	CatBoost Classifier (CatBoostClassifier)
Training Benchmark Dataset	WESAD (Wearable Stress and Affect Detection) Wrist Protocol
Windowing Strategy	30-second Rolling Windows (15s Step, 50% Overlap)
Extracted Feature Vector	23 Biomarkers (NeuroKit2: EDA, HRV, 3-Axis IMU)

Hardware Sampling Frequency	25.0 Hz (ESP32 WROOM-32 USB Serial Telemetry)
Classifier Accuracy (LOSO)	89.72% ± 7.42%
ROC-AUC Benchmark Score	97.52%